Substrate-Field Biology

A Field-Geometry Account of Light, Motion, and Membrane Behavior in Living Systems

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Abstract

Conventional biology treats electromagnetic phenomena in living systems as byproducts of chemistry. This paper argues the reverse: the structured electromagnetic geometry of biological tissue, the substrate field or EMF, is primary. Chemistry does not drive the field; the field sets the geometric conditions under which chemistry occurs. Three phenomena: bioluminescence in marine organisms, coordinated motion in electric eels and schooling fish, and action-potential behavior in plant membranes, are shown to be unified expressions of a single underlying principle: biological systems are field-geometry machines that exploit localized geometry transitions to do work, signal, and self-organize.

1. INTRODUCTION

Biology has done a remarkable job cataloguing what living systems produce. The literature is full of precise descriptions of which ions cross which channels, which enzymes fold which substrates, which signaling molecules bind which receptors with what affinity constants. The data are excellent. The causal story underneath the data is wrong, or at minimum, it is operating at the wrong level of description. The convention is that chemistry drives life. Ions push, proteins pull, ATP fires, and the observable behaviors of organisms are the downstream consequences of these molecular events. This paper argues that the frame is inverted. The substrate field is primary. The geometry of the local electromagnetic environment surrounding and pervading biological tissue sets the conditions under which molecular events occur. Chemistry is not the engine; it is the exhaust.

This is not a mystical claim. It does not require invoking vitalism or any force outside of known physics. Quite the opposite, it requires taking known physics more seriously than biology typically does. Electromagnetism is a field theory. Electrons occupy orbitals whose geometry is determined by the local field. Ion channels are charged protein structures embedded in a lipid dielectric, and their gating behavior is a function of local potential geometry, not a voluntary decision by the protein. The membrane is a capacitor. These are not analogies, they are precise physical descriptions. The argument of substrate-field biology is simply that these physical descriptions should be moved to the front of the explanatory chain, not left as footnotes to the chemistry.

The conventional account of biology at the molecular level is operationally successful in the sense that it generates interventions, drugs that block channels, molecules that inhibit enzymes. But success at intervention does not validate the causal story. Jamming a radio signal by cutting the power line works without knowing anything about electromagnetic wave theory. That does not mean the power line is the correct causal account of the radio broadcast. Biology has, in a very real sense, been cutting power lines and calling it an explanation of the broadcast. The substrate-field framework does not deny that the channels exist, that the enzymes fold, that ATP hydrolyzes. It says that these events are consequences of field-geometry conditions, and that if you want to understand why they happen when and where they do, you need to understand the field geometry first.

Three biological phenomena will be examined in this paper, chosen because they look completely unrelated under the conventional account but become obvious expressions of the same underlying principle once you adopt the field-geometry frame. The first is bioluminescence, specifically the light production of dinoflagellates and jellyfish of the genus *Aequorea*. Conventional biology treats these as chemical reactions: luciferin oxidizes, a photon emerges. That is a description of the output, not an account of what triggered the geometry transition that made the electron drop. The second is coordinated motion — the synchronized discharge of electric eel electrocytes and the collective behavior of schooling fish. Conventional neuroscience treats these as command-and-execute systems with

central controllers issuing signals. They are not. They are field-coherence events, and the mathematics that describes them is not different in kind from the mathematics of coupled electromagnetic oscillators. The third is membrane electrophysiology in plants, the millisecond folding of *Mimosa pudica*, the snap of *Dionaea muscipula*, and the navigational behavior of root tips through soil gradients. Plants have been grudgingly admitted to have action potentials, but the admission has not been followed by a genuine rethinking of what drives those potentials. The field does.

The three phenomena are not chosen to be exhaustive. They are chosen to be triangulating. If you can show that bioluminescence, whole-body motion coordination, and plant mechanosensing all reduce to the same underlying field-geometry mechanism, you have said something about the deep structure of biological systems that the conventional account has missed. That is what this paper tries to do.

2. WHAT IS A SUBSTRATE FIELD

The term "biofield" has a troubled history. It has been borrowed by practitioners of energy medicine, acupuncture theorists, and a range of other traditions that use the word without specifying what the field is, what it is made of, or how it could in principle be measured. This paper is not doing that. The substrate field, as used here, is a precisely defined physical object: it is the structured electromagnetic geometry that pervades and surrounds biological tissue, arising from the spatial distribution and orientation of charged and polar macromolecules within that tissue. It is a real field in the physicist's sense — it has magnitude and direction at every point in space, it carries energy, it propagates changes at the speed of light, and it can in principle be mapped by characterizing the charge distribution and dipole orientation of every macromolecular constituent of the tissue volume in question. The fact that this mapping is technically difficult does not make the field less real; it makes it a hard measurement problem, which is a different issue.

Biological tissue is not electrically neutral at the molecular scale. Proteins carry net charges that vary with pH and post-translational modification. Lipid bilayers are polar structures

with defined dipole moments. Nucleic acids are strongly negatively charged polyelectrolytes. The cytoskeleton, particularly actin and microtubules, carries a net charge distribution that has been shown to be sensitive to mechanical stress and is oriented by cell polarity. When you pack all of these charged, polar, anisotropic molecules together in the precise three-dimensional geometry that constitutes a living cell, the resulting electromagnetic geometry is not random noise. It is structured. It varies by tissue type, by cell cycle stage, by metabolic activity, by mechanical deformation, and by thermal gradient. This structured geometry is the substrate field.

The most important physical structure in the substrate field of animals and plants alike is the biological membrane. The membrane is not correctly described as a barrier with voltage. It is a field structure, a charged dielectric sheet with a defined capacitance, a resting potential maintained by the work of ion pumps, and a geometry that changes in response to mechanical force, temperature, and the binding of charged ligands. The analogy to a waveguide is exact and instructive. A waveguide is not just a tube. The tube is what gives the electromagnetic wave its shape, its mode, its frequency cutoff, its propagation velocity. The wave does not exist independently of the waveguide geometry. In exactly the same way, the chemistry that occurs at and across the membrane does not exist independently of the membrane's field geometry. The channel does not gate because it decided to. It gates because the local potential geometry crossed a threshold that changes the energy landscape of the protein's conformational states. The field geometry determines the chemistry, not the reverse.

Central to the substrate-field account is a mechanism this paper calls the electron drop: a localized, geometry-triggered event in which the field geometry at a specific molecular site crosses a threshold value that makes the current orbital configuration of an electron unstable, causing it to transition to a lower orbital and release energy. This is standard quantum mechanics applied to a biological context. The novelty is not the physics, the novelty is the claim that the threshold crossing is a field-geometry event driven by the macroscopic organization of the tissue, not a stochastic thermal event or a purely chemical

one. When the substrate field at a specific site crosses the threshold, the electron drops. The drop releases a photon in the bioluminescence case, releases heat or drives a bond change in the redox case, or perturbs the local potential and propagates a geometry change to neighboring sites in the membrane-wave case. These are not three separate mechanisms. They are three consequences of the same event type: a field-geometry threshold crossing that triggers a localized electron transition.

This framework makes the membrane the most important organ in biology, which it is — but for the right reasons. The membrane is the primary structure through which the substrate field maintains its geometry, through which geometry changes propagate, and through which the organism transduces interactions with its environment into field-state changes that drive behavior. Every ion channel is a field sensor. Every membrane voltage is a field state. Every action potential is a field wave. The chemistry that biologists measure, the ion concentrations, the protein conformations, the phosphorylation states, are the mechanical correlates of field states, not their causes.

3. BIOLUMINESCENCE AS ELECTRON-DROP GEOMETRY: GLOWING PLANKTON AND JELLYFISH

Start with the ocean at night. A wave breaks and the foam glows blue-green. Charles Darwin described this in his diary from the *Beagle* voyage: the bow wave phosphorescent, the wake milky with cold fire. What he was seeing was dinoflagellates, single-celled marine eukaryotes that emit light when mechanically disturbed. The organism responsible for much of that blue wave glow is *Lingulodinium polyedrum* (formerly *Gonyaulax polyedra*) and related species including *Pyrocystis lunula*, organisms that have been the subject of intensive laboratory study for decades. The conventional account of their bioluminescence goes like this: mechanical agitation of the cell triggers a signal transduction cascade, intracellular calcium levels rise, an action potential opens voltage-gated proton channels in the membrane of organelles called scintillons, the pH inside the scintillon drops to approximately 6, the luciferase enzyme becomes active at this lower pH, luciferin oxidizes,

and a photon is emitted in the 474 nm range. This account is correct in its sequence of events. It is not an explanation of the mechanism. It is a flowchart of the outputs.

What mechanical agitation actually does to the dinoflagellate cell is deform its membrane. The cell wall of *Pyrocystis lunula*, studied with precision by Baran et al. and colleagues at the intersection of biophysics and fluid dynamics, responds to mechanical stimulation in a viscoelastic manner, the light response depends not only on the amplitude of deformation but on its rate. This is not what you expect from a purely chemical trigger. A purely chemical trigger has a threshold concentration: either enough signal molecule is present or it is not. A viscoelastic response is what you expect from a system whose trigger is a mechanical field geometry event, specifically a rate-of-change in membrane deformation that alters the local electromagnetic field geometry at the scintillon membrane faster than the cell's resting geometry can re-equilibrate. What deforming the membrane does, in field-geometry terms, is change the dipole orientation of the charged lipid heads and the embedded proteins at the scintillon surface, which shifts the local electrostatic geometry at the luciferin-binding pocket of the luciferase enzyme. That geometry shift is the threshold event. When the field geometry at the chromophore site of the luciferin molecule crosses the threshold, the highest occupied molecular orbital of the luciferin becomes transiently destabilized, the electron drops to a lower configuration, and the energy difference is emitted as a photon.

The pH change, in this account, is not the cause of the enzyme activation, it is a consequence of the same geometry event. The proton channels open because the local field geometry at the scintillon membrane crossed the gating threshold of those channels. The pH drop follows from the ion movement that follows from the channel opening that follows from the field-geometry threshold crossing. Biology measures the pH because pH is easy to measure. The field geometry that drove all of it is harder to measure, so it gets listed as "the signal transduction cascade" and treated as a black box. The substrate-field framework opens that black box and finds a geometry event inside it.

The capacity analogy is exact and deliberate. A capacitor stores charge across a dielectric. When the voltage across the dielectric exceeds the breakdown field strength, a spark crosses the gap. The cell is a charged dielectric. The scintillon is the gap. The mechanical deformation is the perturbation that pushes the local field over the dielectric threshold. The light flash is the discharge event. This is not a metaphor being applied loosely to biology for rhetorical effect. The mathematics of dielectric breakdown and the mathematics of field-geometry-triggered electron transition are the same mathematics, applied to materials with different dielectric constants and different gap geometries. The dinoflagellate scintillon is a biological dielectric gap, and the bioluminescence flash is a biological spark. The organism evolved this geometry because field-discharge events on the 474 nm scale are detectable by predators and by conspecifics, and being able to trigger them under mechanical stress is adaptive. The organism is not doing chemistry. The organism is exploiting field physics. The chemistry is incidental to the physics.

Now go to *Aequorea Victoria*, the jellyfish that gave molecular biology one of its most important tools. *Aequorea* produces light at its bell margin through the photoprotein aequorin, first isolated and characterized by Osamu Shimomura in the 1960s. Aequorin emits blue light at 470 nm when it binds calcium ions. The green fluorescent protein, GFP, absorbs that blue light and re-emits it as green at 509 nm, giving the jellyfish its characteristic green glow at the tissue level. The 2008 Nobel Prize in Chemistry recognized the discovery and development of GFP, and rightly so, it has been used as a fluorescent reporter in virtually every area of cell biology. But the Nobel-level attention to GFP's utility has somewhat obscured an important question about what is physically happening at the molecular level when aequorin fires and GFP responds.

Conventional biology treats the calcium ion as the signal that triggers aequorin. Calcium binds to aequorin's EF-hand domain, induces a conformational change, the coelenterazine chromophore oxidizes, and a photon is emitted. GFP then "receives" that photon through resonance energy transfer, a process called bioluminescence resonance energy transfer, or BRET, which has been confirmed experimentally and is well-characterized. The substrate-

field account treats the calcium influx differently. The calcium ion, Ca2+, is a divalent charge carrier. When it enters the local volume around the aequorin protein, it does not merely bind in the lock-and-key sense. It reshapes the entire local electromagnetic field geometry around the protein's chromophore. The calcium ion is a field perturbation agent. Its two positive charges reorganize the electrostatic geometry of the binding pocket in a way that crosses the threshold for electron transition in the coelenterazine chromophore. The geometry crosses; the electron drops; the photon is emitted at 470 nm.

What happens next is where the framework becomes particularly clarifying. GFP does not receive the photon because it is chemically "downstream" of aequorin in some signaling cascade. GFP receives the photon because its own chromophore, the 4-(p-hydroxybenzylidene)imidazolidin-5-one formed autocatalytically from three residues in the protein sequence — has a field geometry that is tuned to absorb 470 nm photons. This is antenna physics. An antenna receives a signal not because it is connected to the transmitter by a wire, but because its own geometry is resonant at the transmitted frequency. GFP is a biological antenna, and the fact that it evolved within the same organism as aequorin, in close spatial proximity to aequorin's emission zone, tells you that the organism has solved a resonant field-coupling engineering problem. It built a transmitter and a receiver tuned to the same frequency and deployed them in the same tissue. This is not biochemistry in the conventional sense, it is field-geometry engineering carried out in protein hardware. The organism that evolved this system did not do so because it understood antenna theory. It did so because the physics of field-resonant energy transfer is available to any charged molecular system with the right geometry, and evolution found it.

The broader implication of the bioluminescent systems is this: these organisms did not accidentally develop chemistry that happens to produce light. They specifically evolved geometries that exploit field-threshold transitions at the electron level to produce precise, timed, frequency-specific photon emission events. The frequency is not incidental, it is determined by the geometry of the chromophore. The timing is not incidental, it is

determined by the geometry of the membrane and the threshold of the field-geometry trigger. The organisms are using field physics deliberately, in the engineering sense of that word, to communicate, to deter predators, and in the case of *Aequorea*, possibly to coordinate within-tissue signaling. This tells you that the field is computationally real to these organisms. They are running calculations in field geometry, not in chemistry. The chemistry is the medium, the way a transistor is the medium through which a circuit computation runs, but the computation itself is in the geometry.

4. MOTION AS FIELD-GEOMETRY COORDINATION: ELECTRIC EELS AND SCHOOLING FISH

The electric eel is the obvious starting point for any discussion of biological field use in motion and coordination, because the electric eel has dispensed with any pretense that the field is incidental to its biology. *Electrophorus electricus* and the more recently characterized *Electrophorus varii* and *Electrophorus voltai* generate electric organ discharges, or EODs, from stacks of electrocyte cells that run along most of the animal's body length. Each electrocyte is a flattened, disc-like cell derived from muscle precursors. The posterior face is innervated; the anterior face is not. When the posterior membrane fires an action potential driven by acetylcholine released from the motor neuron terminal, the cell depolarizes asymmetrically, the posterior face depolarizes while the anterior face remains at resting potential, and the two faces of the cell together act as a battery cell with a potential difference of roughly 150 mV. Stack hundreds of these cells in series and you get the high-voltage discharge that can exceed 600 volts in a large specimen of *E. voltai*.

Conventional neuroscience says the eel's nervous system coordinates the electrocytes via action potentials propagating down motor axons. This is true but deeply incomplete. The synchronization of electrocyte discharge across a body-length array of cells must be precise to within milliseconds, or the potentials from individual cells cancel rather than summate. A 2015 study by Markham and colleagues examined the rapid synchronization of electric organ discharge in gymnotiform fish and found that circulating leak currents among the

electrocytes themselves contribute significantly to the synchronization, beyond what motor innervation alone can account for. This is the field doing work. The electrocytes are not passive recipients of nervous system commands that happen to sum correctly. They are participating in a field-coordination event in which the discharge of each cell creates a local field perturbation that propagates to neighboring cells through the surrounding tissue and fluid, and that propagating perturbation contributes to the timing of neighboring cell discharge. The nervous system sets the tempo; the field enforces the synchrony.

The eel does not use its electric organ solely for prey stunning, though that application gets most of the popular attention. It also uses continuous low-voltage discharges for electrolocation, sensing the distortion of its own self-generated field by objects in the environment, and it uses the electric organ for communication with conspecifics. This means the eel is running three field-geometry operations simultaneously: it is emitting a field, reading the perturbation of that field by external objects, and reading perturbations of that field caused by other eels. Each of these operations requires eel's electroreceptors to resolve fine differences in the geometry of the surrounding electromagnetic field. The electroreceptors in the eel's skin are among the most sensitive field-geometry sensors in biology, capable of resolving potential differences in the microvolt range across the skin surface. The eel, in other words, is an animal that exists partly inside its own field and navigates by reading field-geometry distortions. Its field sense is primary to how it experiences its environment, not secondary to vision or chemosensation.

Now move to fish that do not generate conspicuous electric fields, the schooling fish that produce one of the most visually striking examples of coordinated animal behavior. A school of tens of thousands of sardines or herring moves as a coherent object, turning, compressing, and expanding in ways that no individual fish is orchestrating. The conventional ethological account says fish follow simple local rules: maintain a minimum distance from neighbors, align your heading with nearby fish, and be attracted to the center of the group. These rules, run in a computational model, produce schooling-like behavior, so the account has some predictive success. But the mechanism by which each fish

instantaneously knows the position, velocity, and acceleration of its neighbors is not adequately addressed by "they look at each other." Fish in a school react to neighbor movements with latencies too short for visual processing to account for, particularly under low-light conditions.

The mechanism is the lateral line organ: a sensory system running along the fish's body that detects pressure waves and velocity fields in the surrounding water. Each neuromast in the lateral line is a cluster of hair cells capped by a gelatinous cupula that deflects in response to fluid flow. The lateral line does not detect discrete signals from individual neighboring fish. It detects the continuous mechanical field of the water volume surrounding the fish, a field that is shaped by the wakes, bow waves, and body motions of all nearby fish simultaneously. The school moves as a coherent object because each fish is a sensor embedded in a shared mechanical field, and the dynamics of that field couple the fish together without any fish needing to compute a trajectory. This is exactly the mathematics of coupled oscillators in a shared medium. If you have N pendulums mounted on a common beam, they synchronize not because each pendulum "knows" what the others are doing, but because each pendulum is responding to and contributing to the mechanical field of the beam. The school is the same system, scaled up and made of fish instead of pendulums, operating in a fluid medium instead of a rigid beam.

The school is a field mode. It is not a collection of autonomous agents following rules. It is a coherent spatial pattern of motion sustained by the mechanical field coupling of its constituents, the way a standing wave in a resonant cavity is sustained by the geometry of the cavity. When a predator strikes, the school's response, the flash expansion, the fountain maneuver, the split, is a field-geometry response to a sudden large perturbation of the mechanical field, propagating outward from the strike point through the school at a speed determined by the coupling strength of the lateral-line field, not by the visual reaction times of individual fish. This is why fish at the far edge of the school respond before they could possibly have seen the strike. They are not responding to visual information. They are responding to a field wave.

Undulatory locomotion ties the coordination argument back to the body-level field. When an eel, a ray, or a cephalopod moves through water by propagating a wave down its body, it is setting up a traveling wave of mechanical stress through its tissue. That wave has a specific geometry, a wavelength, an amplitude, a propagation velocity, and that geometric wave interacts with the surrounding fluid field to generate thrust by the same fluid-dynamic principles that govern any propulsive wave in a viscous medium. The muscle contractions that execute this wave are the actuators. But the geometry of the wave is primary, it is the pre-existing physical solution to the problem of thrust generation in a fluid, and the animal's motor system implements that geometric solution. Evolution found the optimal wave geometry; the nervous system's job is to run the actuators in the right sequence to produce it. The geometry precedes and constrains the chemistry of muscle contraction, not the reverse.

5. MEMBRANE BEHAVIOR AND PLANT ELECTROPHYSIOLOGY

Plants are where the substrate-field framework has its most dramatic vindication and where conventional biology has most severely undersold the field. The standard story about plant response to stimuli is a chemical story: hormones propagate, stomata respond to CO_2 and abscisic acid, roots grow toward water via auxin redistribution, pathogen responses involve salicylic acid signaling. All of this chemistry happens, and the genes that encode it have been mapped, the protein structures solved, the signaling pathways traced. None of that is in dispute. But the chemical account has a timing problem that has been known for decades and has not been honestly resolved: some of the fastest plant responses simply cannot be explained by the diffusion rates of hormones through tissue. *Mimosa pudica* folds a leaflet in milliseconds after touch. *Dionaea muscipula* snaps its trap closed in under half a second. Hormone diffusion at any measured rate is far too slow to account for the coordination of these events. Biology's answer has been to say "the plant has action potentials", but this answer, if taken seriously, leads directly to the substrate-field framework, and most biologists have been reluctant to follow it there.

If the plant has action potentials, then the plant's membrane is doing exactly what the neuron's membrane does: it is propagating a wave of geometry change along its surface. The membrane depolarizes at the stimulus site because the local field geometry is perturbed by mechanical deformation, just as was described for the dinoflagellate scintillon. That depolarization changes the local field geometry of the neighboring membrane section, crossing its threshold for voltage-gated channel opening, and so on down the tissue. The wave travels at a speed determined by the membrane's capacitance, the local ion conductance, and the resistivity of the cytoplasm and extracellular fluid, the same parameters that determine action potential propagation velocity in neurons. Alexander Volkov and colleagues documented in detail the electrical signal propagation characteristics in *Mimosa pudica*, measuring the velocity of what they called the "alert wave" at approximately 1 cm/s in the pulvinus under flame stimulation, with voltage-gated ion channel blockers interrupting propagation. This is a field wave with measurable speed and defined physical parameters, propagating through plant tissue.

The *Mimosa* folds because a field wave reaches the pulvinus, the motor organ at the base of each leaflet, a tissue specialized for rapid volume change. When the field wave arrives at the pulvinus, it does not trigger a chemical cascade that eventually produces mechanical movement. The wave is already a mechanical event in the membrane, and the mechanical consequence at the pulvinus is immediate: the pulvinus cells, whose membrane potential crosses the threshold for rapid ion efflux, dump potassium and chloride ions out of the cell in a geometry-driven cascade. The water follows osmotically. Turgor pressure collapses. The leaflet folds. The entire sequence from touch to full folding occurs in a time frame that is consistent with field-wave propagation velocity across the leaf, not with hormone diffusion velocity. This is not a subtle distinction, the time scales differ by orders of magnitude. The mechanism is field propagation, full stop.

The Venus flytrap is even more instructive because it has a counting mechanism. *Dionaea muscipula* requires two touches of its trigger hairs within approximately 20 seconds to close the trap, a delay that prevents the plant from closing on raindrops or debris. Armada-

Moreira and colleagues, working at Linköping University and Columbia University, used conformable organic electronic multielectrode arrays to map the action potential propagation in *Dionaea* tissue with spatial resolution that earlier electrode studies could not achieve. They found that the action potential propagates through the tissue with constant speed and without strong directionality, it is a true wave, not a conducted signal limited to particular channels. The two-touch requirement corresponds to a charge accumulation threshold: the first touch fires an action potential that partially depolarizes the trap tissue. The second touch fires another action potential within the time window during which the tissue has not fully repolarized, and the summed field-geometry state crosses the threshold for the motor cell response that drives trap closure. This is field summation. It is the same physics as temporal summation in neurons, and it is operating in a plant that diverged from the last common ancestor of animals somewhere around 1.5 billion years ago.

Root navigation extends the field-geometry framework into the domain of spatial sensing. Roots navigate through soil that is heterogeneous in water content, mineral ion concentration, mechanical hardness, and temperature. The conventional account invokes differential auxin distribution driving differential cell elongation on the two sides of the root tip, the chemotropism account. This happens. But it does not account for the speed or the precision of root navigation in response to rapidly changing soil geometries, nor does it account for the phenomenon of electrotropism, root growth directed by applied electrical fields, which has been documented by multiple groups including Collings and colleagues. Roots navigate electrical field gradients. This is not surprising from a substrate-field perspective: the root tip is a structure with a defined resting membrane potential, and the soil surrounding it is a heterogeneous electrical medium with potential gradients that reflect its composition. The root tip is continuously sampling the relationship between its own internal field geometry and the field geometry of the substrate it is pressing into. Growth occurs in the direction that minimizes field-geometry mismatch between the root's internal state and the external environment, this is resonance seeking, the principle by which any tuned antenna grows toward the field geometry it is designed to receive.

The mycorrhizal network is the final exhibit and perhaps the most remarkable. Mycorrhizal fungi form symbiotic associations with the roots of the vast majority of land plant species, extending networks of fungal hyphae through soil that connect root systems of different plants. These networks are sometimes called the "wood wide web" in popular science writing, with the implication that they are chemical signaling networks through which trees share resources and information. They are that, but they are also, and perhaps more fundamentally, electrical networks. Transmission of electrical signals through mycorrhizal hyphal networks connecting plant roots has been documented by several groups. The signals propagate at velocities consistent with field-wave propagation in membrane systems, not with the diffusion velocities of chemical signals. The network is a biological field network with no central controller, no brain, no nervous system, operating purely on the field-geometry propagation principles that were in operation among living systems hundreds of millions of years before the first neuron evolved. The nervous system did not invent field-wave propagation. It refined and accelerated a mechanism that the rest of biology had been running since the earliest eukaryotes. The substrate field was there first.

6. UNIFYING THE FRAMEWORK

The three phenomena examined in this paper look completely different under the conventional biological account. Bioluminescence is a photochemical reaction driven by enzyme kinetics. Schooling behavior is an emergent property of individually-behaving agents following local rules. Plant action potentials are a curiosity, interesting but somehow less important than plant hormone signaling. Under the substrate-field framework, these three phenomena are three expressions of the same underlying principle at different scales and in different materials, and the unity becomes obvious once you stop treating the chemistry as primary.

Bioluminescence is a field-geometry threshold event: the local electromagnetic geometry at a chromophore site is perturbed by membrane deformation or ion binding, crosses the threshold for spontaneous electron transition, and releases a photon whose frequency is

determined by the geometry of the chromophore, not by the identity of the enzyme. The enzyme is the hardware. The geometry is the program. The dinoflagellate scintillon, the aequorin-calcium interaction, the GFP resonant absorption, all of these are field-geometry events running on protein hardware, the same way a logic gate is a field-geometry event running on semiconductor hardware. The specific protein matters the way the specific transistor material matters: it sets the parameters, but the operation is in the field.

Coordinated motion is a field-coherence event: the synchrony of electric eel electrocytes, the collective behavior of schooling fish, the undulatory wave of the swimming eel, all of these are situations in which a field mode, whether electromagnetic or mechanical, organizes the behavior of a set of coupled physical elements. No central controller is needed because the field is the controller. The nervous system in the eel provides initial conditions and modulates parameters, but the synchronization of thousands of cells into a coherent discharge is a self-organizing field event. The school of fish has no leader because it does not need one, the mechanical field is the leader, and every fish is simultaneously following it and contributing to it.

Membrane electrophysiology in plants is field-wave propagation through a biological dielectric medium. The wave travels because each membrane section, when its field geometry is perturbed beyond threshold, perturbs the next section. The speed is set by the medium's electrical parameters. The biological consequence, folding, snapping, navigating, is the mechanical outcome of cells whose turgor is controlled by the ion distribution that the field wave transiently changes. The field wave is not a metaphor for the biological signaling. The field wave is the biological signaling. Ion channels are field sensors. Membrane voltage is field state. The pulvinus motor is a field-state actuator.

What unites all three is the electron drop: the localized field-geometry threshold crossing that drives a quantum transition with observable macroscopic consequences. In bioluminescence, the drop releases a photon. In redox reactions of mitochondria and chloroplasts, the drop drives electron transfer chains whose energy output powers the cell. In membrane ion channel gating, the drop changes the conformational energy landscape

of a protein, opening or closing a conductance pathway. In every case, the triggering event is a field-geometry condition reaching a threshold, not a chemical concentration reaching a threshold, even though the two are correlated. Concentration affects field geometry; geometry is what actually crosses the threshold. The framework does not erase chemistry, it correctly identifies chemistry as the physical correlate of field-geometry states, rather than the cause of them. Luciferin oxidation does not cause the photon. The geometry event that makes the electron drop causes the photon, and the oxidation is what the luciferin looks like afterward, just as the arc mark on a capacitor plate is what the capacitor looks like after dielectric breakdown. The mark is real; it did not cause the arc.

7. PREDICTIONS AND TESTABLE IMPLICATIONS

A framework without predictions is philosophy. Substrate-field biology makes concrete, falsifiable predictions that differ from the conventional account in ways that are in principle measurable with existing equipment.

First, on bioluminescence: if the photon emission event in dinoflagellate scintillons is a field-geometry threshold crossing rather than a purely chemical rate event, then altering the dielectric properties of the scintillon's local environment without changing luciferin concentration should predictably shift the emission threshold. Specifically, embedding the scintillon membrane in a medium with a different dielectric constant — which can be done experimentally by varying the ionic strength and composition of the bathing solution, or by applying external electric fields, should shift the mechanical force required to trigger bioluminescence in a direction predictable from the dielectric theory. This is a different prediction from the conventional account, which would predict that emission threshold shifts only when luciferin concentration or pH changes. If the threshold shifts with dielectric environment at constant chemistry, the substrate-field account is supported.

Second, on action potential propagation: if membrane action potentials are field-propagation events whose direction and velocity are set by the field geometry of the membrane, then applying an external field geometry that matches the propagating wave

geometry, a traveling-wave electric field applied along the membrane's long axis, phased to match the natural propagation velocity, should be able to accelerate or block propagation in a direction-dependent way. Accelerating a wave traveling in the same direction as the external field, blocking a wave traveling against it. This is not explicable by ion concentration changes alone. If the directional field dependence is confirmed, it is direct evidence for field-wave rather than ion-diffusion propagation.

Third, on root navigation: if root electrotropism is a case of field-resonance seeking rather than simple electrotaxis, then roots should navigate toward artificial field geometries that mimic the field signature of water-rich soil even in the absence of actual water. The field signature of moist soil differs from dry soil in its electrical impedance spectrum, moist soil has a characteristic frequency-dependent conductivity profile. Applying that impedance profile artificially to an otherwise dry soil region, using embedded electrodes running AC signals at the relevant frequencies, should attract root growth toward the artificial field source. The prediction is specific: the response should be frequency-dependent in a way that reflects the root tip's sensitivity to the actual soil-water field profile, not a generic response to any electrical signal.

None of these experiments are exotic. The equipment exists. The predicted outcomes differ clearly from what the conventional account would predict. These are the tests that determine whether substrate-field biology is a genuine advance or a restatement of known physics in unfamiliar language. The prediction is that it is a genuine advance, and the experiments are straightforward enough that there is no good reason they have not been done.

8. CONCLUSION

Biology has been doing physics without admitting it. The membrane is a capacitor, this is in every textbook, stated plainly, and then immediately treated as a metaphor rather than a physical fact. The action potential is a wave, this is also in every textbook, with the propagation velocity calculated from cable theory, and then the field character of the wave

is quietly set aside in favor of discussions about ion concentrations. The bioluminescent flash is a field-triggered electron transition, and yet the literature treats it primarily as an enzyme kinetics problem. The schooling fish is a field mode, and yet the literature models it as a multi-agent simulation. In each case, the physics is present at the surface of the biology, and the biologist looks at it and then looks away.

The reason for this is partly historical, molecular biology's enormous success in the mid-twentieth century established chemistry as the dominant explanatory framework for life, and frameworks that succeed generate practitioners who become invested in maintaining them. It is partly technological, measuring field geometry in living tissue at the molecular scale is hard, and biology evolved in a direction shaped partly by what was measurable. And it is partly conceptual, treating the field as primary requires accepting that the organism is not fundamentally a chemical machine with electrical side effects, but an electromagnetic machine that happens to be implemented in chemistry. That is a large conceptual shift, and large conceptual shifts are uncomfortable.

But the data support it. The timing of plant responses is wrong for chemical diffusion and right for field-wave propagation. The synchronization of electrocyte discharge exceeds what motor innervation alone can produce. The dielectric character of bioluminescent triggering is consistent with a threshold crossing in field geometry. The resonant field coupling between aequorin and GFP is antenna physics, not biochemistry. The mycorrhizal network transmits electrical signals at field-wave velocities through a structure that has no nervous system. All of these point in the same direction.

Substrate-field biology is not a replacement for molecular biology. Molecular biology's descriptions of what the machines are made of are correct and valuable. This framework is the layer underneath those descriptions, the account of why the machines do what they do when they do it. Molecular biology has been standing on that layer without looking down. The electron does what electrons do. The field does what fields do. The organism exploits both, because it is made of matter in an electromagnetic universe, and that is the only universe on offer. It is time to look down.

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